Physics-Informed Machine-Learning Inversion of Chandrayaan-3 RAMBHA-LP Langmuir Probe Sweeps

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The RAMBHA-LP Langmuir probe on the Chandrayaan-3 Vikram lander recorded current–voltage sweeps of the near-surface lunar plasma near the south pole between 24 August and 3 September 2023. Electron density, electron temperature and plasma potential are conventionally extracted by piecewise curve fitting, which fails on 44% of the 29,862 sweeps in the public rawB archive. We trained a physics-informed neural network on 40,000 synthetic sweeps from an orbital-motion-limited forward model, with noise and parameter ranges measured from the real archive, and a loss that also penalises failure to reconstruct the input curve. Applied to the archive, the network recovers 8,343 of the 13,266 classical failures, raising usable coverage from 55.6% to 83.5%. With no ground truth on the Moon, we validate three ways: agreement with an independent classical fit (1.1% in density over 16,596 sweeps), self-consistency between the rising and falling halves of each commanded sweep (2.6%, r = 0.995, 12,418 pairs), and a blind reconstruction of the mission-long density rise from 246 to 493 cm^{-3} by a model never given the date. Median values agree with the published ranges for this instrument, except for density on the first three days. Physics-informed training worsens accuracy on synthetic data yet adds ten percentage points of real-data recovery, showing that simulation alone cannot judge sim-to-real choices. A two-population extension with a working negative control finds no evidence for a second electron population, in agreement with the instrument team, and shows that the standard averaged-sweep diagnostic for multiple populations is not valid evidence on its own.